MEDPRO BILLS

AI Automation Process Design & Implementation Proposal

Current-state workflow • Proposed automation • Integration approach • Technology blueprint

Prepared for: XYZ (company name used as a placeholder)
System: MedPro Bills – in-house claims and document-processing software
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Executive overview

This proposal documents the current patient-document and insurance-claim workflow described by the business, identifies the high-effort manual activities, and defines two integrated automation modules. The objective is to reduce repetitive document preparation and data entry while keeping employees responsible for review, exception handling and final claim submission.

Recommended solution

Build an AI-assisted processing layer alongside MedPro Bills. Module 1 prepares patient-wise folders and document groups before assignment. Module 2 extracts claim data from assigned documents and prepares the Intake Bills form for employee verification. The existing MedPro system remains the system of record.

Document scope

The current-state description is based on the workflow explanation and MedPro Bills screenshots supplied for this discussion. Proposed capabilities and technical choices are recommendations, not confirmation that MedPro currently exposes the required APIs or integration points.

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How to read this proposal

Current process describes activities confirmed in the supplied explanation. **Proposed process** describes the future workflow recommended for automation. **To confirm** marks details that cannot be established from screenshots alone, such as ZIP extraction behaviour, API availability and the internal MedPro technology stack.

01 | Purpose, goals and design principles

The company receives patient and insurance-related documentation, organizes it into patient-specific work packages, assigns the work to employees and processes claims in MedPro Bills. Employees currently spend substantial time reviewing bulk PDFs, separating pages, grouping document types and manually transferring information from documents into structured form fields.

Project objectives

- Reduce manual effort involved in analysing large document batches and preparing patient-wise folders.
- Classify documents and propose page-level tags so employees do not have to identify every page manually.
- Extract patient, insured, insurance, provider, ICD and CPT information from assigned documents.
- Populate the existing Intake Bills form for employee review rather than replacing the existing claim workflow.
- Detect missing, inconsistent or uncertain values before submission.
- Maintain traceability between each extracted value and the source document/page.

Design principles

Principle	What it means in this project
Human-in-the-loop	Employees review uncertain data, confirm patient matches, choose claim status/reason and submit.
MedPro remains authoritative	The existing MedPro Bills application remains the system of record unless the company approves a different architecture.
Traceability	Store source file, page number, extracted value, confidence and reviewer correction.
Fail safely	Do not guess unclear handwriting, medical codes, dates, amounts or patient identity.
Incremental integration	Start with a separate automation service and connect through approved APIs or controlled UI automation.

02 | Current-state process: how work is performed today

The following sequence reflects the process described by the company. Some operational details, particularly the exact behaviour after ZIP upload, still require confirmation.



Current MedPro Bills areas identified in the screenshots

Screen / area	Observed purpose in the described process
Folder Dump	Upload area with a Folders List and Files List. The screen initially prompts the user to select a provider and then a folder.
Assignment	Work allocation area. The screen prompts the user to select a date folder and then a folder/files.
Intake Bills	Claim entry workspace. Patient form fields appear on the left and a PDF viewer appears on the right.

Screen / area	Observed purpose in the described process
PDF tag controls	Dropdown includes document categories such as AOB, Bills, Delivery Receipt, Invoice, Medical Report, Miscellaneous and NF-2. The user described selecting page numbers and saving pages under a tag.
Search	Search form and results grid for locating existing/submitted claim records; visible fields include patient, insurer, provider, claim/policy number, dates, status and CPT amount.
Folder Timeline / Assignment Report / Export	Additional navigation areas are visible. Their detailed operational use has not yet been fully described.

03 | Pain points and automation opportunities

Current manual activity	Why it consumes time / creates risk	Improvement opportunity
Reviewing bulk PDFs	Large batches contain multiple patients and document types; page boundaries may be unclear.	OCR, page analysis, patient identification and document classification.
Splitting and folder creation	Employees manually separate pages, create patient folders and organize files locally.	Automated page segmentation, patient-wise grouping and folder generation.
ZIP preparation and upload	Repeated packaging and upload steps add handling time.	Automated ZIP creation and approved Folder Dump integration.
Page tagging in Intake Bills	Employee selects a tag and identifies page numbers for each document category.	AI-suggested page-to-tag mapping with employee approval.
Reading and entering form data	Employees switch attention between PDF and form and retype information.	Structured extraction and field-level auto-fill.
Handwritten ICD/CPT details	Handwriting may be difficult to interpret; incorrect coding or amounts can affect claim quality.	Handwriting-capable OCR, reference validation and mandatory review for low-confidence values.
Existing patient checks	Employees may need to compare new details with prior records.	Candidate matching using multiple identifiers, with employee confirmation.
Missing information review	Incomplete or inconsistent fields may only be noticed during or after entry.	Pre-submission completeness and consistency checks.

Key design decision

Automate preparation and data transfer, not clinical or claim judgement. The system should suggest values and statuses, but employees must resolve ambiguous identity matches, unclear codes and missing information and remain responsible for final submission.

04 | Module 1 – AI Document Processing & Folder Automation

Objective: Automate the work performed before employee claim processing: ingest source documents, identify patients, split and classify pages, prepare patient-wise folders and make approved work packages available in MedPro Bills.

Proposed workflow



Detailed functional capabilities

- **Google Drive ingestion:** connect to an authorized Drive location, select or detect incoming batches, preserve original filenames and record the source file ID.
- **PDF analysis:** inspect page text and images; apply OCR to scanned pages; detect blank, rotated, low-resolution or unreadable pages.
- **Patient identification:** extract candidate patient name, date of birth, SSN or other approved identifiers, claim/policy clues and page-level evidence.

- **Patient grouping:** associate pages using multiple identifiers and document context. Do not group solely on a name where identity is uncertain.
- **Document classification:** predict the MedPro tag/category for each page or page range, including AOB, Bills, Delivery Receipt, Invoice, Medical Report, Miscellaneous and NF-2.
- **PDF splitting:** produce patient-specific PDFs and, where appropriate, separate category PDFs while preserving page order and original source references.
- **Folder and ZIP generation:** create the agreed folder naming convention, manifest and ZIP package for each batch/provider/date.
- **Folder Dump integration:** upload approved output and record the MedPro upload/processing confirmation. If direct integration is unavailable, provide a controlled employee upload step.

Illustrative output structure

Example only – confirm naming conventions with operations

Provider / Intake Date / Batch ID → Patient folder → AOB.pdf, Bills.pdf, Invoice.pdf, Medical_Report.pdf, NF2.pdf, Miscellaneous.pdf. Keep a manifest listing source PDF, patient candidate, output PDF, source page numbers, predicted tag, confidence and review status.

Confidence and exception handling

Classification outcome	System action
High confidence + consistent identifiers	Prepare patient grouping and document category automatically; include in review sample or approved batch flow.
Medium confidence	Place the page/group in an employee review queue; show evidence and alternative patient/category suggestions.
Low confidence / conflicting identifiers	Do not auto-group. Flag for manual decision and retain the original page.
Unreadable / damaged page	Mark as unreadable or low quality; request rescanning or manual review.

05 | Module 2 – AI Intake Bills & Claims Automation

Objective: Reduce manual data entry after work has been assigned. When an employee opens a patient case in Intake Bills, the automation reads the assigned documents, extracts structured values, maps them to the existing form and presents a populated draft for verification.

Proposed employee experience



Field extraction and mapping scope

MedPro form section	Candidate data to extract / map	Important control
Patient Information	Initial, first name, last name, SSN, DOB, gender	Validate date formats and patient identity; avoid guessing missing values.
Address	Address lines, ZIP, state, county, city, address type	Normalize formats but preserve source value and flag conflicts.
Contact	Contact type, phone number, extension	Validate number format; do not infer missing digits.

MedPro form section	Candidate data to extract / map	Important control
Insured Information	Policyholder identity and contact details	Do not assume insured = patient; use the form's same-as-patient option only when supported by source evidence.
Insurance Information	Insurer, policy number, claim number, date of accident (DOA)	Use insurer master data and source evidence; flag conflicting policy/claim numbers.
ICD Code	Treating/signing provider, ICD code and description; multiple diagnoses	Validate code/description against approved reference data and applicable date/version.
CPT Code	Serving address, CPT code, amount, description, DOS start/end	Verify handwritten code, amount and service dates; allow multiple CPT entries.
Claim Status / Reason	Possible suggested status or reason based on completeness	Employee selects/confirms; never submit based solely on an AI suggestion.

Existing patient matching

The extraction service can compare the newly extracted patient details with existing MedPro records, if an authorized search API or integration is available. Matching should use a weighted combination of approved identifiers (for example, DOB plus name and a permitted unique identifier), show candidate records and require employee confirmation. A name-only match must not automatically merge or overwrite records.

Document tag and page suggestions

The same classification results from Module 1 can be reused in Intake Bills. If the incoming case includes additional documents, Module 2 can analyse those pages and propose the MedPro tag plus page range. The employee previews and approves the mapping before the system saves it. This avoids running the same OCR/classification work twice where the results are still valid.

06 | Target end-to-end workflow and MedPro integration

The two modules should share one processing pipeline and a common case/batch identifier. Module 1 prepares the documents; Module 2 consumes the approved patient package when an employee opens the assigned case.



Recommended integration pattern

Integration point	Preferred approach	Fallback if no API exists
Google Drive → automation	Google Drive API with least-privilege access and batch tracking.	Employee uploads/selects files through the automation dashboard.
Automation → Folder Dump	MedPro-supported API/service endpoint for folder/file upload and status confirmation.	Authorized browser automation or guided upload with employee confirmation.
Assignment → Module 2	Use case/folder identifiers from MedPro API or event/webhook to retrieve the assigned case.	Employee launches Auto-Fill from the case and selects the relevant documents.

Integration point	Preferred approach	Fallback if no API exists
Module 2 → Intake form	Backend/API integration using validated field mapping and MedPro business rules.	Playwright-based UI interaction, with selectors and checks maintained against UI changes.
Tag/page grouping	Call MedPro document-tagging service with page ranges and tag IDs.	Populate the UI tag controls and save only after employee approval.

Integration prerequisite

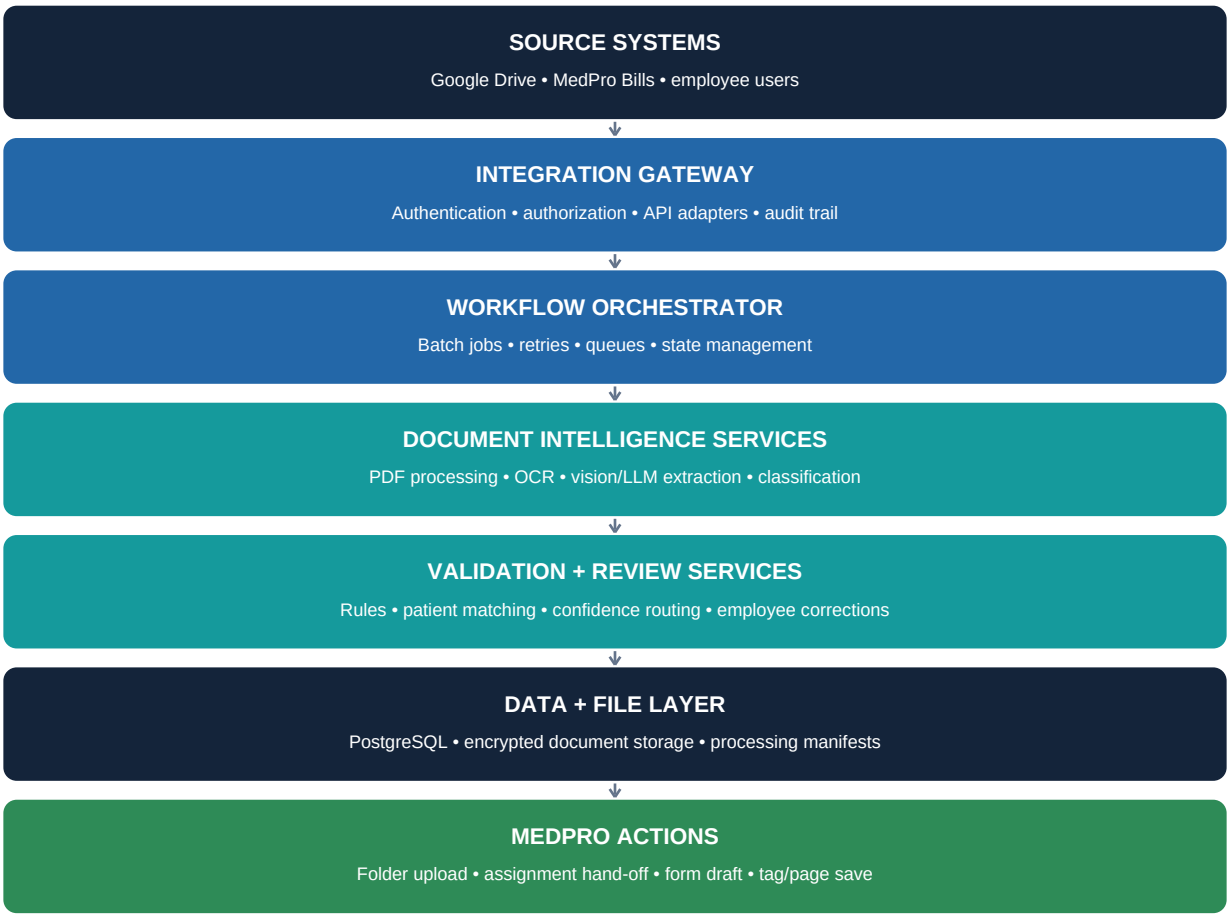
Before implementation, the MedPro owner must confirm available APIs, authentication, upload endpoints, tag/page save operations, patient search and claim-save interfaces. Do not write directly to the production database or bypass MedPro validation rules without explicit authorization.

07 | Technology stack and system architecture

The recommended stack is designed to keep document processing, AI extraction and MedPro integration modular. The final selection should be confirmed against the company's hosting, security and existing MedPro architecture.

Layer	Recommended technology	Purpose / notes
Employee automation dashboard	React + Vite + TypeScript + Tailwind CSS	Batch status, review queue, page preview, extraction results, corrections and approvals.
Backend API	Python + FastAPI + Pydantic	Secure APIs, orchestration, validation schemas and MedPro integration adapters.
Background processing	Celery + Redis (or managed queue)	Run long PDF/OCR/AI jobs asynchronously; retries, job status and failure handling.
PDF processing	PyMuPDF + pypdf	Render pages, extract embedded text, split/merge PDFs, preserve page references.
OCR / document AI	Azure AI Document Intelligence or another approved enterprise document AI provider	Printed text, layout/table extraction and supported handwriting recognition; validate language and document performance.
LLM / vision reasoning	Approved vision-capable model with structured JSON output	Classify document types, reconcile context and extract structured fields. Use only an approved healthcare-compliant deployment.
Validation	Pydantic + Python rules + approved code/reference datasets	Field formats, required fields, code-description checks, date/amount consistency and confidence routing.
Database	PostgreSQL	Batches, documents, page maps, extracted fields, confidence, corrections, audit logs and job status.
File storage	Encrypted object storage / approved company storage	Original PDFs, page images, split files and generated ZIPs; retention policy required.
Drive integration	Google Drive API	Authorized ingestion, file metadata, download and source tracking.
MedPro integration	Existing MedPro API/backend; Playwright only if required and approved	Upload folders, retrieve assignments, populate fields and save tag/page mappings.
Deployment / monitoring	Docker; approved cloud or company infrastructure; centralized logs and monitoring	Repeatable deployment, secrets management, health checks, alerts and auditability.

Logical architecture


SOURCE SYSTEMS
Google Drive • MedPro Bills • employee users
INTEGRATION GATEWAY
Authentication • authorization • API adapters • audit trail
WORKFLOW ORCHESTRATOR
Batch jobs • retries • queues • state management
DOCUMENT INTELLIGENCE SERVICES
PDF processing • OCR • vision/LLM extraction • classification
VALIDATION + REVIEW SERVICES
Rules • patient matching • confidence routing • employee corrections
DATA + FILE LAYER
PostgreSQL • encrypted document storage • processing manifests
MEDPRO ACTIONS
Folder upload • assignment hand-off • form draft • tag/page save

08 | Data quality, security and human review controls

Patient and insurance records can contain sensitive personal and health information. The automation must be designed as a controlled healthcare-data workflow, not as a general-purpose public AI upload tool.

Required safeguards

- **HIPAA and vendor review:** assess applicable HIPAA obligations, execute required Business Associate Agreements (BAAs) with vendors handling protected health information, and confirm the service configuration is eligible for the intended use.
- **Data minimization:** send only the pages and fields needed for the task; avoid using real patient data in development, demos or prompt testing.
- **Access control:** role-based permissions for operators, reviewers and administrators; least-privilege access to Drive, storage and MedPro.
- **Encryption:** encrypt data in transit and at rest; manage keys and secrets securely.
- **Auditability:** log who uploaded, processed, reviewed, corrected, approved and submitted each case, including timestamps and source references.
- **Retention and deletion:** define approved retention periods for originals, page images, extracted text, temporary files and ZIP packages; securely delete temporary copies when no longer needed.
- **Model and prompt controls:** prevent document content from being treated as instructions; restrict extraction to approved schemas and avoid unnecessary free-form output.
- **Human approval:** require review of uncertain patient matches, handwritten ICD/CPT codes, amounts, dates, conflicts and claim status before submission.

Data quality controls

Control	Expected behaviour
Field-level provenance	Every extracted field stores source filename, page number and, where possible, bounding-box/location evidence.
Confidence thresholds	Set thresholds per field type after testing; do not assume one threshold works for patient identity, handwritten codes and ordinary printed text.
Conflict detection	If two pages show different DOBs, policy numbers or claim numbers, flag the conflict instead of selecting one silently.
Duplicate detection	Identify repeated files/pages and possible duplicate patient cases; retain originals and ask for review where uncertain.
Code validation	Use current, approved ICD/CPT reference datasets and date-aware rules; AI output alone is not proof that a code is correct.
Employee correction feedback	Record corrections for quality measurement and controlled improvement; do not automatically retrain models on PHI.

Important limitation

OCR and AI can misread poor scans, handwriting, abbreviations and overlapping documents. The system should expose uncertainty and source evidence. It must not invent missing values or autonomously make medical, coverage or claim adjudication decisions.

09 | Implementation roadmap, deliverables and success measures

A phased rollout reduces operational risk and allows the team to measure accuracy before enabling more automation.

Phase	Scope	Key deliverables / exit criteria
0. Discovery & access	Map real workflows; inspect MedPro API/source access; agree folder naming, tag IDs, required fields, security and test data.	Approved process map; field mapping; integration feasibility report; test plan; data-handling approval.
1. Module 1 – MVP	Upload/select Drive batch; OCR; classify; patient grouping; manual review; split and generate folders/ZIP.	Test batches processed end-to-end; page traceability; reviewer approval; no unreviewed uncertain grouping.
2. Folder Dump integration	Connect approved upload interface; upload approved packages; reconcile success/failure and duplicates.	Successful test uploads; clear retry handling; upload audit trail; assignment hand-off confirmed.
3. Module 2 – MVP	Intake case document selection; extract patient/insurance/provider/ICD/CPT fields; display mapped draft and evidence.	Field-level review UI; correction workflow; validation results; no automatic submission.
4. Tagging + patient match	Suggest page tags/ranges; retrieve candidate existing patient records; employee confirms.	Verified tag/page save process; match quality measured; duplicate prevention checks.
5. Pilot + rollout	Run in parallel with existing process; measure accuracy, time and exception rates; train employees; expand gradually.	Pilot sign-off; SOPs; monitoring; rollback plan; production approval.

Suggested success measures

Metric	How to measure it
Document preparation time	Average employee minutes per batch before vs. after automation.
Page classification accuracy	Correct document category / total reviewed pages, measured on a representative labelled sample.
Patient grouping accuracy	Correct patient-to-page assignments / total reviewed page assignments.
Extraction accuracy	Exact-match or field-specific accuracy for each field type, reported separately for printed and handwritten data.
Employee correction rate	Percentage of AI-populated fields changed by employees, by field type.
Straight-through preparation rate	Percentage of pages/cases prepared without manual correction, only after quality thresholds are approved.
Claim preparation time	Average employee time from opening an assigned case to a verified ready-to-submit draft.
Exception and rework rate	Cases returned for missing/incorrect documents or data, compared with baseline.

Pilot approach

Start with representative historical or synthetic/de-identified test batches that include multi-patient PDFs, mixed document types, low-quality scans, handwritten ICD/CPT information, duplicate names and missing fields. Compare AI output with employee-verified ground truth. Set acceptance thresholds with operations and compliance before production use.

10 | Assumptions, dependencies and decisions to confirm

The following items are not established by the supplied screenshots and process explanation. They should be confirmed with the operations lead and MedPro technical owner before final estimation or implementation.

Question / dependency	Why it matters	Owner to confirm
Does Folder Dump automatically extract the uploaded ZIP, or are further manual steps required?	Determines whether the automation should create one ZIP per batch, per patient, or a different package structure.	Operations / MedPro owner
Are source documents one large multi-patient PDF, multiple PDFs, or both?	Affects ingestion, page-boundary detection and patient grouping strategy.	Operations
How are patient folders named today?	Required for consistent folder generation and MedPro compatibility.	Operations
Are document tags and page ranges saved per PDF, per patient or per case?	Defines the page-tag mapping schema and integration action.	MedPro owner
Is there an API for Folder Dump, Assignment, patient search, Intake Bills save and tag/page grouping?	Determines the safest integration path and avoids fragile UI automation.	MedPro technical owner
Can the automation access MedPro source code or backend services?	May enable a supported integration adapter and reuse of existing validation rules.	MedPro technical owner
What fields are mandatory for each claim type / insurer?	Required to build accurate validation rules and prevent false missing-field alerts.	Claims operations
Which ICD/CPT reference version and coding rules are used?	Required for date-aware code validation and description matching.	Claims/compliance team
Which OCR/AI vendors are approved for PHI and what BAAs/security controls are in place?	Required before any real patient data is sent to a third-party service.	Security / compliance
What are the rules for claim status and status reason?	AI may suggest completeness issues, but the employee must select the correct operational status.	Claims operations

Recommended decisions

- Approve the two-module scope as the target solution.
- Keep employee approval before Folder Dump upload where grouping confidence is not high.
- Keep employee verification and final submission mandatory in Module 2.
- Prioritize API/backend integration; use browser automation only when supported integration endpoints are unavailable and the MedPro owner authorizes it.
- Run a discovery and pilot phase before committing to production accuracy, processing time or cost targets.

Target outcome

A smoother workflow in which bulk documents are automatically analysed and organized before assignment, and assigned claim documents are converted into a verified Intake Bills draft. Employees spend less time sorting pages and retyping data, and more time reviewing exceptions and completing the claim accurately.

Appendix | MedPro screen elements referenced

The supplied screenshots show the Folder Dump Folders List / Files List, Assignment Folders List / Files List, Intake Bills patient and address fields, contact and insurance sections, ICD and CPT sections, claim status and status reason dropdowns, PDF viewer tag dropdown, and Search results. The proposal uses these visible elements to define the automation scope. No patient-identifying information from the screenshots is reproduced in this document.

End of document